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The Paradox of Time Experience in the Digital Age and Its Roots

ABSTRACT

This study based on the perspective of the history of philosophy to systematically demonstrate the essential identity between time and consciousness, revealing the essence of time as the basic form of human perception, experience, and cognitive world. Building on this theoretical foundation and incorporating insights from critical social theory—particularly the critique of social acceleration—to analyze temporal experience in the digital age. Our research finds that the digital technology revolution is causing unprecedented temporal alienation: the dialectical coexistence of ephemerality and permanence, acceleration compulsions and experiential scarcity, busyness and meaninglessness. Through critical analysis, we identify the deep-rooted source of this temporal paradox, that is, the dual differentiation of the nature of time brought about by technological acceleration, which detaches time from the structure of consciousness, leading to the dual split of individual time and social time, and causing individual time to be completely subordinated to the social time system. Through this analysis, the study advances the critique of social acceleration theory by establishing its deeper philosophical foundations.

KEYWORDS

time, consciousness, identity, paradox, binary differentiation

Introduction

Time is the eternal mystery of humanity. From the primitive tribes looking up at the stars in awe, to modern atomic clocks cutting seconds with precision; from Augustine's philosophical thought that time is "tension of consciousness itself." (Augustine, 1997, p.306), to Einstein's reconfiguration of the spacetime continuum in relativity, time has always been a fundamental dimension intertwined with human existence. In the age of digital capitalism, this entanglement takes on unprecedented contradictory forms: technological revolution pushes time measurement to the quantum level of precision, yet individual experiences of time are plunged into unprecedented chaos; digital media promises freedom in time management, yet the temporal discipline of technical has infiltrated the biological rhythms of neurons; humanity's narrative of conquering natural time reaches its peak, yet the sense of emptiness is devouring the integrity of life. This eerie reality calls for a reexamination of the essence of time.

Based on this, this article seeks to trace the conceptual history of time in Western philosophy, reinterpret its essence, and reveal the intrinsic identity between time and consciousness. Methodologically, it employs conceptual analysis within the history of philosophy to systematically elucidate the inherent connection between time and consciousness. Building on this foundation, the study integrates critical social theory, particularly the critical theory of social acceleration, to conduct a critical analysis of temporal experience in the digital age. While Rosa's incisive analysis illuminates the paradox between technological acceleration and the scarcity of time, the philosophical roots of this paradox have not been sufficiently explored. This paper aims to address this gap by situating critical social theory within the broader perspective of the history of philosophy. It argues that the underlying cause of this paradox lies in the dissolution of the identity between time and consciousness induced by technological acceleration, which consequently leads to a dichotomous fragmentation of the nature of time.

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1. The essence identity of time and consciousness: A philosophical historical examination

"Time" and "consciousness" are perennial themes in philosophical discourse. The philosophical inquiry into consciousness boasts a long, intricate history, with its conceptualization evolving significantly across different thinkers and eras. In ancient Greece, no concept equivalent to the modern understanding of "consciousness" existed. Philosophers instead debated the "soul". Plato regarded the soul as an immortal, rational essence governing the body; Aristotle, meanwhile, saw the soul as the "form" of a living body—its actuality—emphasizing the identity of soul and body. "The soul, being the form of a natural body that potentially possesses life, is necessarily substance, and this substance is actuality."(Aristotle, 2016, p.31). During this period, "consciousness" was largely understood as a cognitive function subordinate to a grander cosmic or theological framework, lacking an independent subjective status.Augustine internalized this external "soul" into the human "mind" and further equated the mind with memory, asserting that "Mind and memory, however, are one and the same" (Augustine,1997, p.306).

This inward turn was further reinforced by Descartes, who, through his method of radical doubt encapsulated in "I think, therefore I am," established "thinking" as the indubitable foundation of certainty. Consciousness here was equated with a thinking substance— "a thing that doubts, understands, affirms, denies, is willing, is unwilling, and also imagines and has sensory perceptions." (Descartes,1986, p.27)—thereby establishing the principle of subjectivity.Following Descartes, Locke provided a clearer definition of consciousness, arguing that thought is inseparable from consciousness, which he defined as the "perception" of everything that occurs in one's own mind. He was the first to explicitly link personal identity to the continuity of consciousness, proposing that consciousness constitutes personal identity. "Consciousness always accompanies the present sensations and perceptions, and it is only by consciousness that everyone is to himself that which he calls self. " (Locke, 1983, pp. 309-310).

Although Descartes, Locke, and other modern philosophers established consciousness as a subjective principle, they largely treated it as a passive "blank slate" receiving stimuli from the objective world. Kant elevated consciousness from this passive receptor to an active constructor. His concept of "transcendental apperception" refers to that self-consciousness which can accompany all representations and synthesize them into objects that are "mine". "It is that self-consciousness which, because it produces the representation 'I think' that must be able to accompany all others and which is one and the same in all consciousness, cannot itself be accompanied by any further representation." (Kant, 2013, p.103). Thus, consciousness was no longer a mere mirror of the world but the very condition of possibility for the world to appear to us. Time, in turn, became the inner form of consciousness, the necessary condition for transcendental apperception to synthesize all representations. Building precisely upon this foundation, Husserl proposed that consciousness itself is a temporal flow, constituted by the interplay of "primal impression" (the now), "retention" (the just-past held in grasp), and "protention" (anticipation of the imminent future), forming a living field of experience (Husserl, 2009, pp. 61-65).

Aristotle was the first philosopher to explicitly articulate the intrinsic relationship between time and consciousness(i.e. soul). Though he defined time as "the number of motion with respect to before and after" (Aristotle, 1982，p.125), he saw this connection between time and things as based on human sensations or consciousness. Because when we say that time is about the number of motions of things, it is always premised

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on human “sensation” of this motion; without “sensation” , without human “consciousness” , time would not exist. “If nothing but consciousness or the rational part of consciousness can perform the action of counting besides anything else, then if consciousness does not exist, neither can time” (Aristotle, 1982， p.136).

It is on this basis that Augustine further proposed the theory of “mind (consciousness) time”. He opposed Aristotle’s view of time as the number of motion. In his view, time is nothing but an “tension” . but what exactly is it an tension of ? He believed that “I would be very surprised if it is not tension of consciousness itself.” (Augustine, 1997, p.306). Therefore, time can only be the extension of thought, it does not exist in any actual space, but only in our minds. Time is simply the “impression” left in our minds by things passing through the “present”. According to this viewpoint, each of us has our own time, in which we pursue the purpose of life and accumulate “impressions”. Due to the differences in “impressions”, our perceptions of time are completely different. What others want to forget as the “past” may be exactly what I am looking forward to in the “future”; what others are experiencing as the “present” may be a vivid memory of the “past” for me, and so on. Time is constantly generated by the three stages of human thought working: “The mind expects, and attends, and remembers”. (Augustine, 1997, p.309)

With the advancement of natural sciences, reason finally shattered the constraints of medieval theology and began to pursue the truths of the natural world with immense fervor. As noted by process philosopher Alfred North Whitehead: “At the close of the Middle Ages, a new direction of thought emerged. Inventions stimulated thinking, which in turn accelerated progress in the observation of nature, while Greek manuscripts revealed the discoveries of antiquity.…By 1700, when Newton completed his monumental Mathematical Principles of Natural Philosophy, the world had entered a brand-new modern era.”(Whitehead, 2010, p.9) Since that, the theme of philosophy has evolved into exploring and explaining nature, becoming what is often referred to as “natural philosophy” in that historical period. Demonstrating the existence and nature of time and space from the perspective of scientific experiments has become one of the main tasks of modern natural philosophy. Therefore, Isaac Newton’s absolute view of time and space has become the main representative of modern natural philosophy. He equated time with space as a container enveloping the world, with the movement of matter progressing along the track of time. As a result, the meaning of life and free will are both nullified in such an objective time, where time becomes a completely external absolute existence to life and motion, and a divisible quantity. Although Newton’s absolute view of time has its limitations and faced criticism and opposition from Leibniz and others, it is a product of the development of modern natural science and a significant embodiment of technological rational logic, namely the theoretical premise of clock time.

However, as the founder of German classical metaphysics, Kant explicitly rejected the Newtonian concept of absolute time. He was not concerned with whether time exists objectively or in what manner it subsists, but rather pursued further questions: What is the meaning of time for human beings? How does the reality of time manifest itself to us? Consequently, Kant endowed time with reality, making it the foundation for the subject’s apprehension of objective appearances and a fundamental precondition for sensuous cognition. This achieved a shift in the conception of time from ontology to epistemology. In his view, time is merely a form of inner sense, a self-moving form of intuition in our minds’ perceptual world. It governs the relationships of various representations within our minds. Because these relationships have no shape and do not occupy any space, we can only use analogy to represent their shape. That is, representing the sequence of time by an infinitely extended line. Therefore, “Time is only the subjective condition of our (human) intuition (it is at any time sensuous, that is, if we are stimulated by an object) and as such, nothing in itself outside the subject” (Kant, 2013, p.55).

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Kant's strict correspondence of time and space to the dualistic cognitive paths of "inner sense" and "outer sense" was opposed by Hegel. He first discussed time and space as a "unity", considering time to be space. In Hegel's view, a purely "affirmative" thing is lifeless, and it must transition to its opposite. Space is an affirmative existence that must negate itself. Therefore, space is essentially an "inner negation", and its truth is the "self-overcoming of its various moments", which is actually time. However, Hegel emphasizes not the temporality as "change" itself, but as the abstract form of this "change", that is, as the conceptual time. Because the essence of spirit is the concept. "The concept is the power that governs time, time is merely the exteriority of this negativity. Only natural things, due to their finitude, are subject to time; while real things, that is, ideas, spirit, are eternal" (Hegel, 1980, p.49). Thus, Hegel elucidates the essential connection between time and spirit. On the one hand, as concepts, time and spirit are identical; on the other hand, as intuitive change, time is the developmental process of spirit's self-realization, it is the "exclusivity" of the spirit, where spirit "enters" into time.

Although Hegel's concept of "spirit" has an essential difference from "consciousness," it still ultimately points to a kind of metaphysics of time. It can be seen that the entire history of Western philosophy interprets the essence of time, always linking it to human subjective experience, consciousness, or mind. Either time is seen as the extension of consciousness itself, or time is seen as the basic form of perceiving and experiencing the world. It is within this tradition that Bergson further defines time as the "extension of consciousness", where our joys and sorrows are different forms of the extension of time: joy is a state of consciousness towards the future, while sorrow is a state of consciousness towards the past, which is a impoverishment of our sensations and concepts (Bergson, 1958, p.10). Similarly, Heidegger proposes that "temporality is the meaning of being-here" (Heidegger, 2012, p.8). These views collectively reveal the essence of time, where time is both the form of human perception of the world and the dynamic process of self-realization.

However, the arrival of the digital age has brought about an unprecedented paradox in the experience of time: on the one hand, technological acceleration gives humans the illusion of controlling time, as if time has truly become a form of self-realization of consciousness; on the other hand, individuals are deeply trapped in the dilemma of time scarcity and meaninglessness, struggling to find themselves in that seemingly controllable time.

2. The triple paradox of time experience in the digital age

The digital age usually refers to a historical stage characterized by the widespread application of digital technology, which is a new stage of human social development following the industrial age. Digital technology (including but not limited to 5G communication, Internet of Things, artificial intelligence, blockchain, and quantum computing) not only realizes the demediation of information, but also reconstructs the production, distribution, and consumption logic of information through algorithms and network protocols. All of this "fundamentally changed the way we are born, live, learn, work, produce, consume, dream, strive, or die" (Castells, 2001, p.37). When the French poet Baudelaire pointed out that the painters of modern life expect to "extract from poetry the content of its historical background, to extract the eternal from the transitory" (Baudelaire, 2014, p.14), he already foresaw the new time paradox brought about by the industrial age. As humanity transitions from the industrial age to the digital age, the modern time paradoxes vaguely depicted by Baudelaire gradually become universalized and deepen in the accelerating rule of the Internet, the precise segmentation of algorithms, and the temporal discipline of power, presenting as a triple time paradox of ephemerality and permanence, acceleration and scarcity, busyness and meaninglessness.

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We adopt the term “paradox” to emphasize the seemingly contradictory, counterintuitive, yet coexisting realities of temporal experience in the digital age, which constitute systemic internal tensions beyond mere “ambiguities”. While “dialectics” reveals their contradictory unity, “paradox” more directly captures their acute contradictory nature for individual experience. Furthermore, the proposed triad of paradoxes elucidates distinct dimensions of these temporal contradictions: the ephemerality-permanence paradox manifests in the tension between infinite digital storage and rapid informational obsolescence; the acceleration-scarcity paradox reflects how the intense compression of time coincides with growing subjective experiences of time poverty; and the busyness-meaninglessness paradox reveals how high-intensity temporal saturation fails to yield corresponding sense of value or satisfaction. These three interwoven paradoxes collectively constitute the distinctive temporal predicament of the digital era, entrapping individuals within an illusion of technological empowerment while plunging them into profound states of loss of control and anxiety.

2.1 Ephemerality and Permanence

This is an era without “permanence” , yet people’s hearts are filled with a longing for permanence. Artists with their acute intuition, were the first among the bustling crowds to grasp the new characteristic of the modern world: “transience”, yet they so eagerly hoped to capture eternity within it. The paradox lies in how their way of capturing is to continuously depict images of “passing” “momentary” “flowing” which are non-eternal. They view the transformation of modern experiences as the fleeting of time and the temporary storage of space, and consider the accompanying events as arbitrary and incidental. (Frisby, 2013, p.29) Just as Marx and Engels revealed in the *Manifesto of the Communist Party*, “Constant revolutionising of production, uninterrupted disturbance of all social conditions, everlasting uncertainty and agitation distinguish the bourgeois epoch from all earlier ones” (Marx, Engels, 2012, p.403).

With the information technology revolution, we have transitioned from the “hardware” era to the “software” or digital era. The “instability” and “changeability” described by Marx in time experience are becoming increasingly universal.What artists once experienced as a distinctive tension between the eternal and the transient is now deeply embedded in the structural transformation wrought by digital technology itself, becoming a widespread paradox of temporal experience in the digital age. The persistence of modern society, paradoxically, depends on its capacity for constant disruption and change. We therefore witness a continuous, disruptive restructuring of time and space—a shift from “hardware” to “software”, from a “heavy” to a “light” modernity.

The term “hardware” is borrowed from Zygmunt Bauman’s *Liquid Modernity*, where he uses computing metaphors to describe modernity changing experience of time and space. In his view, pre-modern life was characterized by a unified understanding. Distances in time and space (“very far”, “a long time”) were understood in relation to the capabilities of the human body and animal power—what might be termed “wetware”. Modern innovations in transport and communication shattered this unity, leading to a decisive differentiation and a transition from “hardware” to “software”. Time itself became a primary target for mastery. The strategy involved constructing massive, fixed, and heavy infrastructures—base stations, ports, railways, airplanes,cars—in an attempt to solidify and control the fleeting nature of time. As Bauman notes, “Time has become the problem of the “hardware” humans could invent, build, appropriate, use and control, not of the hopelessly unstretchable “wetware” nor the notoriously capricious and whimsical powers of wind or water indifferent to human manipulation.” (Bauman, 2000, p.111).

However, the “software” world that has developed with the advancement of the internet, artificial

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intelligence, etc., has completely eliminated people's control and ownership of time. Time has become "instantaneous", becoming immediate, right away, and even now. " we are witnessing the initial outcomes of a "generalized arrival", in which everything that arrives has no need to depart.…… The journey thus loses its successive components and focuses solely on arrival."（Virilio, 2004, p.22）Not only is it difficult for people to "hold on to" or "control" "a period of time", but time itself has also lost its meaning. The distance between people is no longer the speed of trains, cars, ships, or even planes, but the speed of telephones, text messages, WeChat, and emails; work no longer follows the conventional, rationalized time, the kind of large factory-style lifelong contract or agreement disappears in a series of digital time acceleration forms such as gig work, short-term work, outsourcing, and transfers. Moreover, in order to "control" time, we even change our biological sleep patterns, turning sleep into the most unnecessary restriction, and the "24/7" world of round-the-clock operation completely dissolves the physiological constraints that humans have had for thousands of years. However, as Jonathan Crary put it: "24/7 is a time of indifference, against which the fragility of human life is increasingly inadequate and within which sleep has no necessity or inevitability." (Crary, 2013, p.9)

This is the first paradox of the time experience in the digital age: inventions such as the Internet and quantum communication are meant to facilitate "instant" (a form of eternity) communication between people, yet they bring about a constant change in all forms of communication; the emergence of artificial intelligence gives people hope for "immortality" (a form of eternal life), but it increasingly virtualizes people's experience of time; the continuous innovation of digital medical technology has finally overcome the problems of "short life" and "early death" that traditional medicine couldn't solve in the past, yet the real phenomena of "short life" and "early death" continue to increase unabated, with terms like "death from overwork" "heart attack" "stroke" "suicide" and "depression" filling the lives of this generation.

2.2 Acceleration and Scarcity

In the experience of modern time where everything is constantly passing by, there is a deeper logic of time, which is everything is accelerating, namely the "its constitutive experience is that of a monstrous acceleration of the world, of life, and of each individual's stream of experience." (Rosa, 2013, p.35). The speed of our washing, eating, doing household chores, commuting, working, studying, even sleeping, and dating is constantly increasing. Therefore, Rosa proposes three types of understanding social acceleration: technical acceleration, social change acceleration, and the acceleration of the pace of life, to present the temporal acceleration landscape of the digital age.

First, technical acceleration. This is the most obvious and influential form of acceleration in modern society. It is a "the intentional, technical, and above all technological (i.e., machine-based) acceleration of goal-directed processes . Paradigmatic examples are processes of transportation, communication, and production (of goods and services)." (Rosa, 2013, p.71). This form of acceleration is the most intuitive and easily measurable and proven. Taking the transportation industry as an example, the history of the development of human movement speed from pre-modern to contemporary society has experienced from walking to horseback, from steamboats to trains, automobile, planes, and even spacecraft. Human mobility speed has increased from four to five kilometers per hour to several hundred or even thousands of kilometers per hour. The direct consequence of this form of acceleration is the dramatic compression of physical space, while the acceleration of information and digital technology has, to a certain extent, eliminated the meaning of physical space.

Next, the acceleration of social change. This means the faster replacement of social structures (social, political, legal, cultural, managerial, etc.) and the acceleration of society itself. In other words, in the process of

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social change, “the rates of change themselves change , i.e., they accelerate.” (Rosa, 2013, p.74). Taking technological innovation as an example, it involves not only the speed of technological innovation itself, but also the speed at which society accepts new technologies, which in turn confirms a process of societal acceleration. For instance, from the invention of the radio at the end of the 19th century to achieving 50 million listeners, it took society 38 years; while achieving the same scale of demand for television only took 13 years; from the first internet user to the 50 millionth user took 4 years, but from the birth of Deepseek to reaching one billion users only took a mere week. These data all demonstrate a form of societal acceleration, where the speed of significant changes in societal practices and behavioral orientations is increasing.

Finally, the acceleration of the pace of life. The acceleration of technological innovation and social change ultimately leads to the acceleration of the pace of life. But why is this? Because fundamentally, the results of the first two accelerations should bring about the liberation of the living world, allowing people more time to enjoy life. However, the reality is just the opposite, as we experience the ever-increasing pace of life. For example, we must constantly strive to shorten the time for watching movies, attending gatherings, dating, etc. This is precisely the opposite side of modern social time acceleration that Rosa perceptively identified, namely the present contraction (Gegenwartsschrumpfung) and the scarcity of time.

In the former case, Rosa pointed out that social acceleration necessarily means the passing away of past acceleration and the arrival of future acceleration, resulting in the present becoming increasingly “contraction”. Social acceleration of change means “an increase of the rate of decay of action-orienting experiences and expectations and as a contraction of the time periods that determine the present of respective functional, value, and action spheres.” (Rosa, 2013, p.76). Social acceleration of change or “contraction of the present” applies to the entire system, culture, and all practices and cognitions of politics, professions, technology, aesthetics, and so on. For example, the speed of changes in family structures, work or profession, friends’ phone numbers, office hours, turnover of political parties, etc. is gradually accelerating compared to the past. This is what causes the acceleration of modern life pace, not deceleration.

As for the latter, “the acceleration of the pace of life” means “an increase in the quantity of actions or experiences within a certain time unit. That is to say, this is because of the desire or the feeling of the necessity to do more things in less time” (Rosa, 2018, p.21). Specifically, it involves two levels: subjectively, due to the increase in the amount of information and tasks brought about by technological acceleration and social acceleration, people feel an unprecedented sense of urgency and pressure. We worry about not meeting deadlines, not understanding new “memes” on the internet, missing the next subway train, being stuck in traffic, and so on. Objectively, there are two ways to measure the acceleration of the pace of life: the first is to measure the “shortening” of the time spent on actions, such as measuring the time spent on eating, sleeping, walking, leisure activities, family conversations, etc. A large amount of evidence shows that the time we spend eating, sleeping, and being with our family is significantly less than the previous generation. The second is to measure the “compression” of the time spent on actions and experiences. For example, within a certain period, by reducing rest or interval time to do more or experience more things, or by doing more things simultaneously, such as cooking, watching TV, and talking on the phone at the same time.

As shown in the figure 1, the change between technical acceleration, the acceleration of social change, the acceleration of the pace of life, and the past-present-future triple structure of time forms a triangular pattern.

This is the second paradox of modern time experience: as everything accelerates, we do not feel an increase in time, but instead increasingly feel that time is insufficient. “I don't have time” has almost become a common

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reason for contemporary people to refuse additional life experiences. Acceleration does not bring an increase in the amount of time, but instead a shortage and scarcity of time. However, when we constantly sacrifice individual life time due to “time scarcity” and busily rush without a moment's rest, we find it difficult to feel the meaning of life in this endless busyness.

2.3 Busyness and Meaninglessness

Does time have meaning? Obviously, in terms of physical time or astronomical time, time is unified, homogenous, and flows at a constant speed without ever stopping. However, today, people do feel the acceleration of time passing, experiencing the scarcity and lack of time. So, where does the meaning of time come from? Who gives meaning to time? It is only humans who can give meaning to time. And what is this meaning? We believe it is the freedom and satisfaction that each individual feels in time. If a person cannot have a moment of freedom and satisfaction throughout their life, then we will admit that their life is meaningless. It is for this reason that Friedrich Nietzsche calls on people to “become who you are” in the powerful law of history .

However, how to become “yourself”? Who defines “yourself”? For example, today, when a person wants to follow his heart and live a pure personal life, what will he do? Besides personal life activities such as eating and sleeping, he will inevitably seek out things that he deems meaningful to do. Once his personal time meaning must be realized through others, through society, he has to embed his own time rhythm into the accelerated law of social time. The result is that people can only achieve self-worth by constantly being busy, by constantly shortening their own life time. They even change their own time rhythm according to the social time rhythm, sacrificing individual time for social time. This is also the root cause of the global phenomenon of death from overwork: “The fear of being stuck in place in a high-speed society can lead to cultural diseases such as apathy, depression, world-weariness, neurasthenia, or various forms of stress today. ... Because the changes and dynamics in individual life or the social world (i.e. individual or collective history) are no longer meaningful, directional development factors.” (Rosa, 2018, p.54).

This reveals the third paradox of the time experience in the digital age: technology acceleration has not brought ease and satisfaction to modern people, but endless confusion. When people are busy continuously according to the social time rules, they do not achieve the meaning of life in this busyness, but instead it is endless fatigue and emptiness. The relentless endeavors results in a world of “the loss meaning”,“obscurity of purpose”, and “the loss freedom” (Taylor, 2020, p.31).

So, what is the root cause of this time paradox of experience? Why hasn’t technological progress brought liberation and leisure to humanity? This requires us to further explore the issue of time differentiation brought about by technological progress.

3. The deep roots of the time experience paradox: the dual differentiation of the essence of time.

Technology and its spawned industry constitute one of the most significant drivers of progress and achievement in modern society, having dramatically reduced material scarcity for many, though not all, populations globally. However, the industrial assembly lines, developed transportation industry, precise digital algorithms created by technology have greatly surpassed its “natural attributes”, causing humans to not only be unable to control the “magic” it summons from within, but also suffer from its backlash, leading to a complete

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reversal of their own lives. This consequence is first manifested in humans' experience of time: time is no longer an entity aligned with consciousness, but becomes split from individual consciousness and becomes a product of social construction. Time is dichotomized into individual time and social time, with individual time completely subordinate to social time, to the extent that time is no longer a form of perception of individual consciousness, but a measure of social production and development time.

Social time is " the time of convergency and divergency of movements of the total social phenomena, whether the total social phenomena are global, group or micro social and whether or not they are expressed in the social structure." (Gurvitch, 1964, p.27). The overall social phenomenon is a kind of real, holistic interrelation and influence formed between people (including political, economic, cultural, etc.) structure. In modern society where capitalist mode of production dominates, the role and status of social time are more evident than ever before. It is a complete system of machine time that transcends natural time: minutes, scales, cycles, calendars, etc. This kind of time replaces the dispersion, disorder, and contingency of individual time with unity, precision, and stability. We "do not pass it as a steady and standard order composed of the ticking sound of a clock, but mainly as an order composed of chunks of time containing some predictable, and widely repeated social events." (Lewis, Weigert, 1981)

Individual time becomes the carrier of social time. Individuals must embed both physiological time and psychological time into social time in order to obtain their own social survival time (working time, labor time, part-time work time, etc., labor time for survival) and life time (time for individuals to satisfy their own needs and development through social ways such as learning, consumption, entertainment, etc.). As shown in the figure 2 below:

First of all, according to human natural attributes or physiological characteristics, everyone has their own physiological time experience. A newborn baby, for example, their physiological time experience is only about eating and sleeping, because their individual time has not yet entered the social time. They can immediately fall asleep when they feel sleepy, or wake up immediately when they feel hungry, without being disturbed by any social time rhythm. "People experience the rhythms of their inner activities, such as heartbeat, breathing, menstrual cycle, hunger, rest, sleep cycles, and the rhythms of natural phenomena cyclically, entering into the awareness of time... Each person's constantly repeated habits, such as finding food, working, resting, and sleeping, are woven into the automatic forms of consciousness, expressing not only a person's sense of time, but also helping with their internal and external adaptations. Biological time is manifested in the most stable and general habits related to the functions of the body."(Mukerjee, 2009, p.34)

Secondly, psychological or conscious time is a unique form of time that distinguishes humans from other organisms. The experience of this type of time depends entirely on an individual's feelings, emotions, interests, etc. For example, when a person feels happy, their personal time flows faster; conversely, when a person feels bored, dull, or upset, their personal time flows slower. "Generally speaking, time filled with change and interesting experiences feels short as it is lived, but when we look back on it, it feels long. On the other hand, time with few experiences feels long as we live it, but seems short in retrospect." (James, 1919, p.624) Therefore, each individual's experience of personal time is completely different, largely depending on the individual's own physical and mental feelings.

Finally, social time is completely different. It is not based on individual time, but on the level of social development and the sustainable breadth as its goal, making individual time completely subordinate to itself. In such time, "there are repeated social events and activity rhythms superimposed on individual unique experiences

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of biological and psychological continuity, and this rhythm determines the time calculation in society. The foundation for individuals to adapt to society cannot be unreliable personal time experiences, but must be time that is invariant and common to all individuals, only on this basis, cooperation in economic and social activities is possible" (Mukerjee, 2009, p.35). On one hand, society tells us when to eat, sleep, engage in activities, drink, rest, etc., social time completely controls our physiological time; on the other hand, psychological time is clearly influenced by a social experience, our psychological experience of time depends on a series of experiences in social time. The psychological time of personal experiences from learning to university and eventually becoming labor force always passes quickly, while the time experience after retirement and returning home is much slower, "because the elderly individuals clearly experience processes that are less valued, or processes that are less important in the social sense." (Lewis, Weigert, 1981).

Therefore, individual time and social time have generated contradictions and conflicts, which are increasingly deepened under the reinforcement of the laws of technology and industry. The contradictions between the finiteness of individual time and the infiniteness of social time, the disorderliness of individual time and the precision of social time, the scarcity of individual time and the richness of social time, etc., force individuals to seek the meaning of time by integrating into social time, further intensifying the dominance of social time.

This is the true root of the paradox of technological acceleration and time scarcity revealed by Rosa. Because the premise of acceleration we set is that our total amount of tasks (events) remains the same, that is, doing the same things in the same amount of time does not increase. However, the fact is that while technology makes the experience of time faster, it also increases the volume of social affairs. "The growth rate of affairs exceeds the rate of technological acceleration, therefore, in the face of technological acceleration, time continues to become more scarce" (Rosa, 2018, p.28). Emails are always filled with messages to reply to, and phones are full of information that cannot be finished, all of which take up more of our time than in the past letter-writing era. Similarly, the inventions of washing machines, cars, etc., not only did not liberate people's time accordingly, but instead reduced or shortened it. "Our previous scope of life may only have covered a few kilometers, but now we travel thousands of kilometers for work or tourism, driving or flying. And now we change clothes every day, unlike a century ago when we may only wash our clothes once a month (or longer)." (Rosa, 2018, p.27)

Not only that, technology has also created such a rich and colorful world, so rich that everyone wants to experience "infinite" social time as much as possible with their "limited" individual time. Therefore, everyone rushes to join the whirlpool of social time, even at the expense of changing or sacrificing personal time. Unfortunately, our short life is far from being able to fully experience all things in this world. So, the acceleration of the pace of life becomes an effective means for us to reduce the gap between individual time and social time. "Acceleration thus becomes a strategy for eliminating the difference between world time and our life time" (Rosa, 2018, p.37) . Little do we know that this acceleration only further deepens the dual split between individual time and social time, which in turn leads people to constantly sacrifice personal time to chase social time, entering an infinite cycle of time closure.

As shown in the figure 3:

In summary, while Rosa accurately describes the paradox between technological acceleration and time scarcity, he fails to uncover its underlying philosophical roots. Building on this foundation, our study deepens his theoretical framework by pointing out that the crux of the paradox lies in the dissolution of the identity between time and consciousness induced by technological acceleration, which consequently leads to a

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dichotomous fragmentation of time into social time and individual time. By integrating the critique of social acceleration with the context of the history of philosophy, our analysis not only clarifies the philosophical essence of this paradox but also further reveals two additional dimensions of the paradox—the tension between the ephemeral and the eternal, and that between busyness and meaninglessness. Thereby, it enriches and expands Rosa's theoretical framework.

4. Conclusion

The paradox of time experience in the digital age originates from the dual differentiation of time and consciousness brought about by technological acceleration, dividing time into individual time and social time. In this context, social time constructs and governs individual time, with individual time completely subordinated to social time. In the industrial age, people need to follow the rhythm of social time, such as a week, nine to five, to arrange their own time, leading to the common consequence that people's self-time only belongs to weekends or after five on weekdays. With the advent of the digital age, people further lose their self-time, being filled with short videos, social information, uninterrupted news, etc., during after work, weekends, and even times for sleeping, washing up, and using the restroom. Therefore, the fundamental approach to transcending the paradox of temporal experience in the digital age lies in breaking the dilemma of temporal fragmentation caused by technological acceleration and restoring the essential identity between time and consciousness. This pathway will constitute the central issue to be explored in our subsequent article.

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References:

Aristotle. (2016). *The complete works of Aristotle (Vol. 3)*. M. Litian, Ed. Beijing: China Renmin University Press.

Aristotle. (1982). *Physics.* Z. Zhang, Trans. Beijing: The Commercial Press.

Augustine, S. (1997). The *Confessions*. Maria Boulding, O.S.B., trans. New City Press.

Bergson, H.(1958). *Time and Free Will* . F.L.Pogson,M.A., trans. Dover Publications,Inc.

Baudelaire, C. (2014). *The Painter of Modern Life*. X. Hu, Trans. Beijing: China Translation & Publishing Corporation.

Bauman, Z. (2000). *Liquid Modernity*. Cambridge: Polity Press.

Castells, M. (2001). *The Rise of the Network Society*. Z. Xia, Trans. Beijing: Social Sciences Academic Press.

Crary, J. (2013). *24/7 :Late Capitalism and the Ends of Sleep*. London • New York: Verso.

Descartes, R. (1986). *Meditations on first philosophy*. J. Pang, Trans. Beijing: The Commercial Press.

Frisby, D. (2013). *The Fragments of Modernity: Theories of Modernity in the Works of Simmel, Kracauer and Benjamin*. H. Lu et al., Trans. Beijing: The Commercial Press.

Gurvitch, G. (1964). *The Spectrum of Social Time*. Myrtle Korenbaum, M. Korenbaum, Trans. & Ed., with the assistance of P. Bosserman. Dordrecht-Holland: D. Reidel Publishing Company.

Husserl, E. (2009). *On the phenomenology of the consciousness of internal time*. L. Ni, Trans. Beijing: The Commercial Press.

Hegel, G. W. F. (1980). *Philosophy of Nature*. L. Zhixue & H. Xue, Trans. Beijing: The Commercial Press..

Heidegger, M. (2012). *Being and Time*. J. Chen & Q. Wang, Trans. Beijing: SDX Joint Publishing Company.

James, W. (1919). *Principles of Psychology*. New York: Henry Holt &.Co.

Kant, I. (2013). *Collected Works (Vol. 3)* , Q. Li, Trans. Beijing: China Renmin University Press.

Locke, J. (1983). *An essay concerning human understanding*. W. Guan, Trans. Beijing: The Commercial Press.

Lewis, J. D. and Weigart, A. J. (1981) *the Structures and Meanings of Social Time*. Social Forces, 60. https://doi.org/10.1007/978-1-349-20869-2_7.

ARTICLE IN PRESS

Marx, K., & Engels, F. (2012). *Selected works (Vol. 1)*. Beijing: People's Publishing House.

Mukerjee, R. (2009) . *Time, Technics and Society*, from *The Sociology of Time*. edited by John Hassard. H. Wen Zhu & J. Li, Trans. Beijing: Beijing Normal University Press.

Rosa, H. (2013). *Social acceleration: a new theory of modernity*. J. Trejo-Mathys, Trans. New York: Columbia University Press..

Rosa, H. (2018). *Beschleunigung und Entfremdung: Entwurf einer Kritischen Theorie spätmoderner Zeitlichkeit*. Lu Dong, trans. Shanghai: People's Publishing House.

Taylor,C. (2020). *The Malaise of Modernity*. L. Cheng, Trans. Nanjing: Nanjing University Press.

Virilio, P. (2004). *Speed of Liberation* .Y. Lu, Trans.Nanjing: Jiangsu People's Publishing House.

Whitehead, A. N. (2010). *Science and the Modern World*. Q. He, Trans. Beijing: The Commercial Press.

Figures legends

Figure 1. The triangular relationship between technical acceleration, acceleration of social change, acceleration of the pace of life, and the temporal structure of past, present, and future. Technical acceleration drives the acceleration of social change, which in turn accelerates the pace of life, leading to the accelerated contraction of the present and the accelerated disappearance of the past and advent of the future.

(Source: Authors' elaboration based on Rosa, 2018)

Figure 2. The dual structure of time. Individual time (comprising biological time and psychological time) is embedded within and determined by social time. Social time governs both biological rhythms (e.g., sleep, eating) and psychological experiences of duration.

(Source: Authors' elaboration)

Figure 3. The vicious cycle of temporal alienation in the digital age. Technological acceleration leads to the dichotomy between individual time and social time, compelling individuals to chase social time, which further reinforces technological acceleration and deepens the split.

(Source: Authors' elaboration)

Data Availability

This manuscript is a theoretical and philosophical analysis. No empirical datasets, code, or survey instruments were generated or analyzed during the current study. All materials and sources supporting the arguments and findings are fully documented and available through the cited references.

Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

Ethics Statement

This article does not contain any studies with human participants performed by any of the authors.

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Author contributions

LR and JX contributed to the conception of the study. LR wrote the manuscript， JX revised the manuscript. All authors contributed to the article and approved the submitted version.

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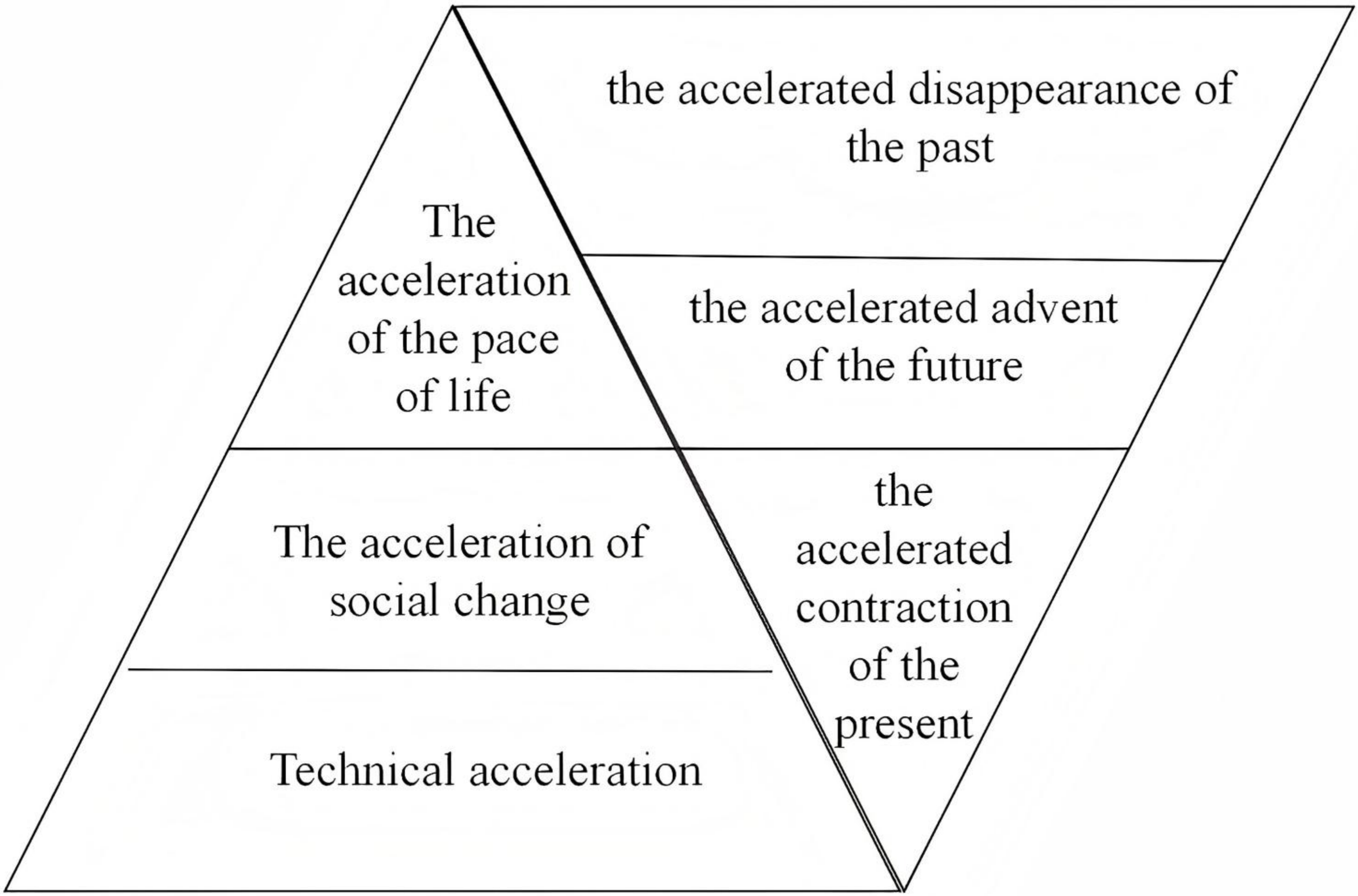
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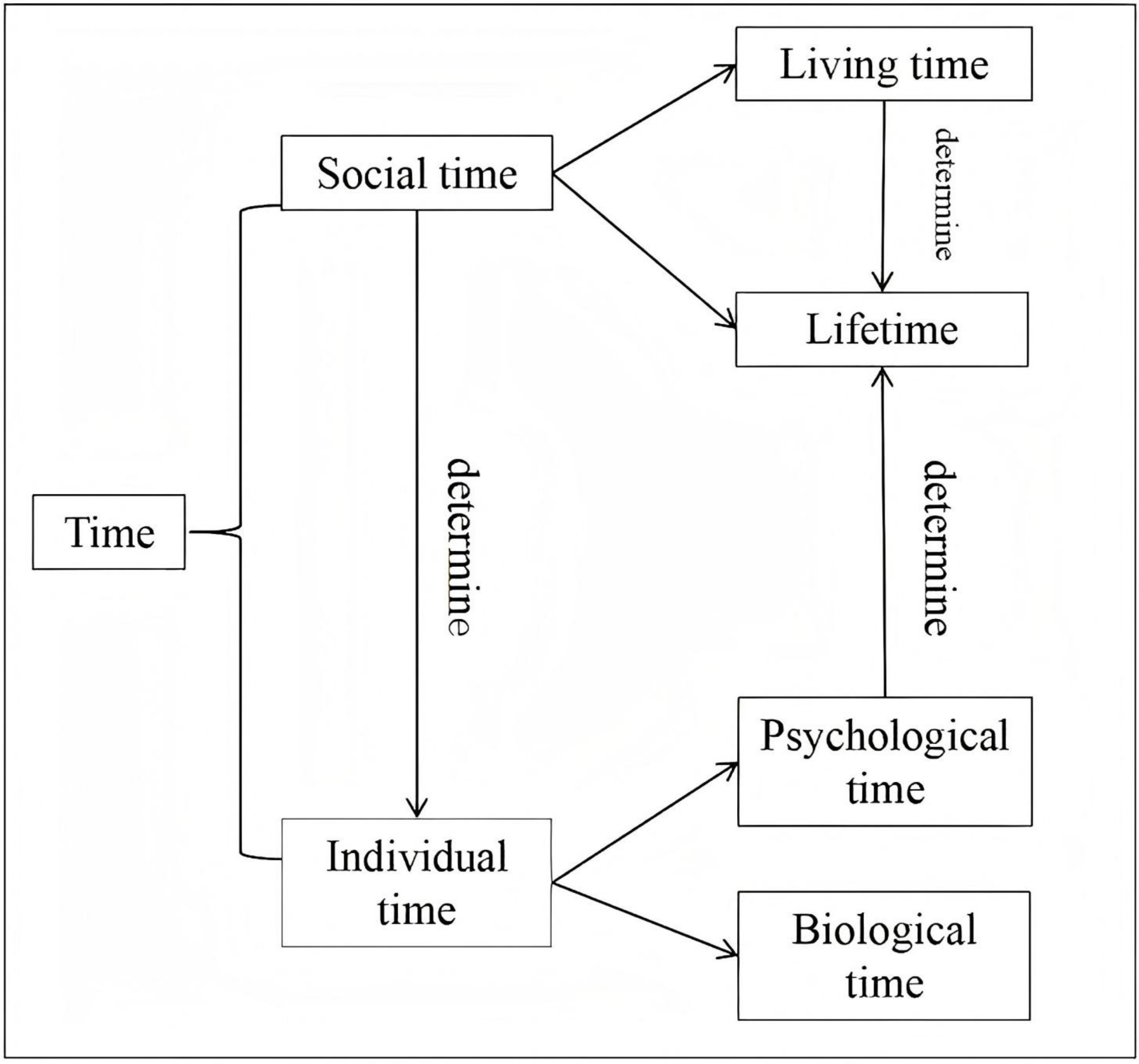
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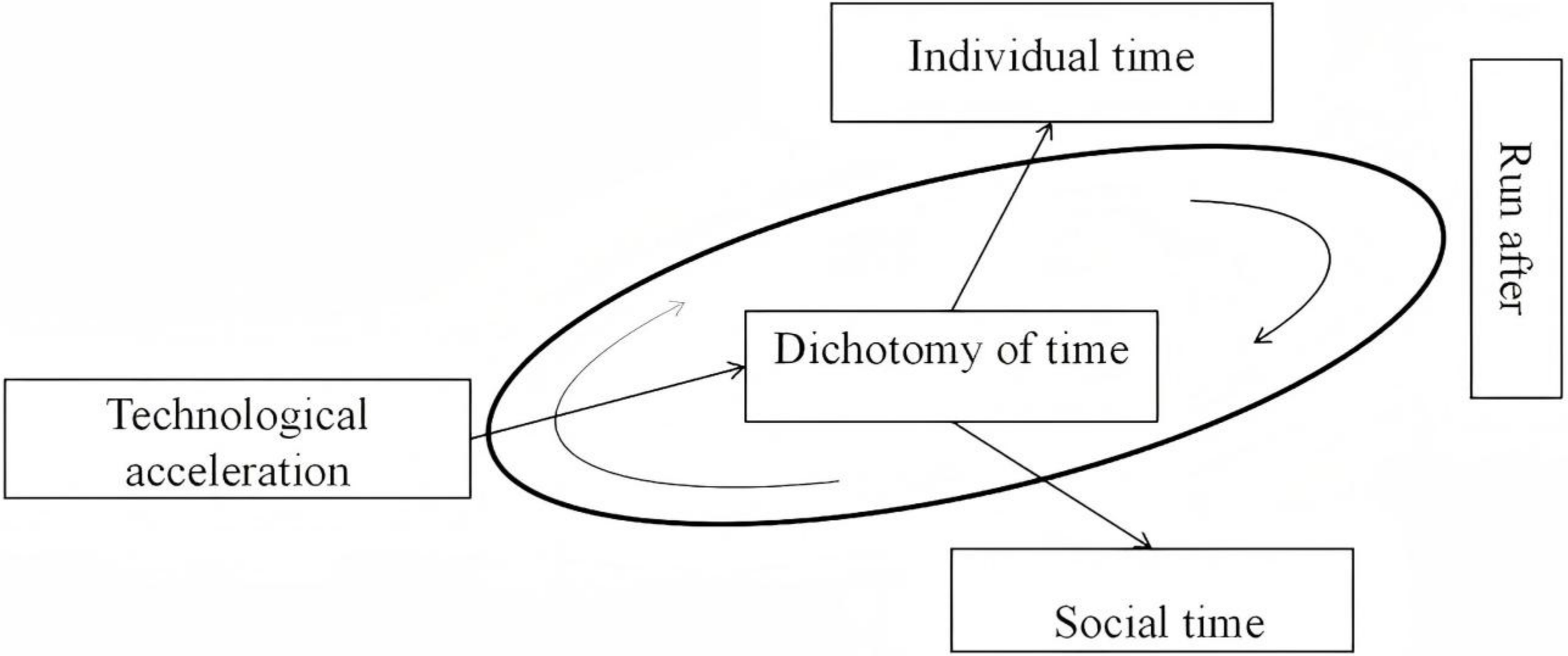
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